Major Project Presentation - Group No. 6

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Identification of Medicinal Plants Using Image Processing

Automating accurate, scalable medicinal plant identification through transfer learning with CNN architectures — Xception, MobileNetV2, and ResNet50.

Why Accurate Plant Identification Matters

Traditional Importance

Medicinal plants are central to Ayurveda and modern pharma, rich in active biochemicals used in treatments.

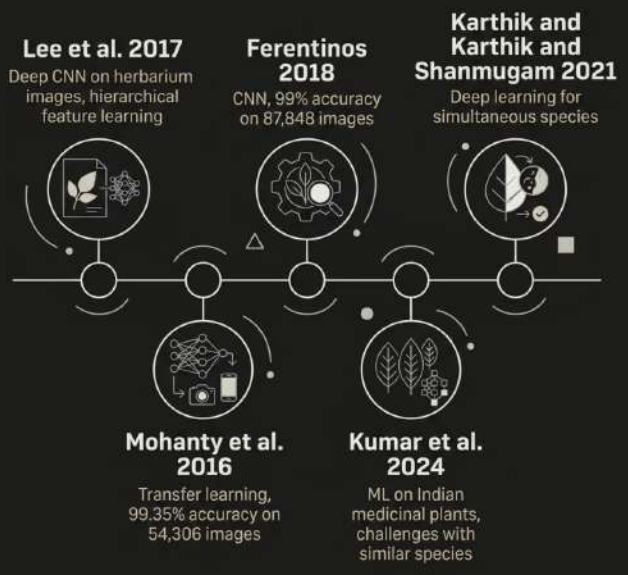
Risk of Misidentification

Wrong identification leads to reduced treatment efficacy or harmful allergic and toxic responses.

Limitations of Manual Methods

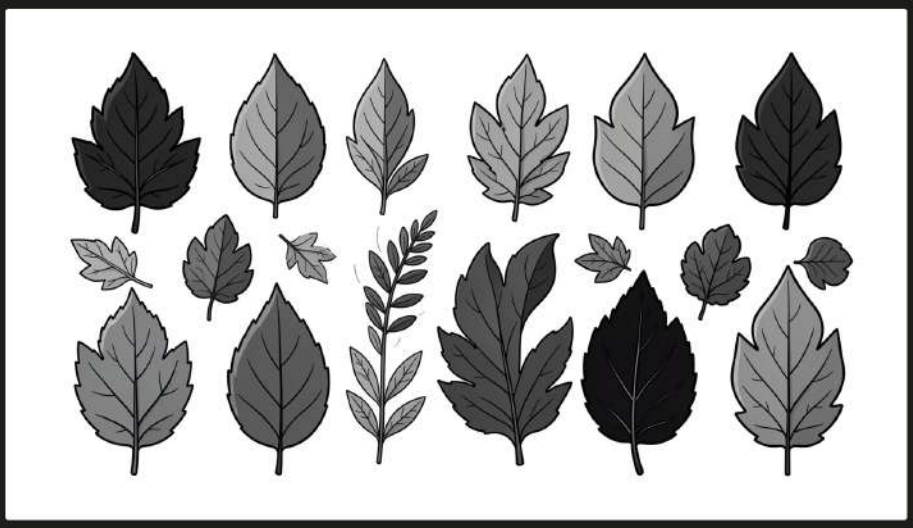
Traditional botanical identification is time-consuming, expertise-heavy, and prone to human error.

Literature Review: Key Prior Works



Prior research demonstrates CNNs excel at plant classification and disease detection. Transfer learning consistently improves performance on limited datasets. Studies on Indian medicinal plants highlight challenges in distinguishing morphologically similar species — the core problem this work addresses.

Dataset & Preprocessing



Indian Medicinal Leaves Dataset (Kaggle)

- **5,945 high-resolution images** across 80 medicinal plant classes
- Species include Tulsi, Aloe Vera, Neem, Betel, and others
- ~74 images per class, varying lighting and backgrounds
- Split: 80% training, 10% testing, 10% validation
- Resized to 299×299 px, pixel-normalized to [0,1]
- Augmentation: random flip, zoom, and rotation

Methodology: Three CNN Architectures

Xception

Depthwise separable convolutions; 22.9M params; 299×299 input. Best at fine-grained leaf pattern extraction.

MobileNetV2

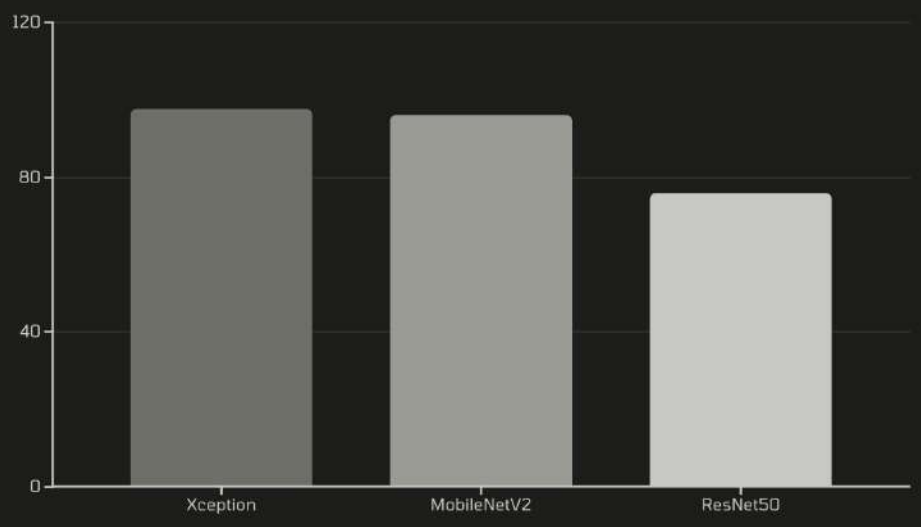
Inverted residuals and linear bottlenecks; 3.5M params; 224×224 input. Optimized for edge/mobile deployment.

ResNet50

Residual skip connections; 25.6M params; 224×224 input. Deep architecture tested for complex feature learning.

All models used ImageNet pre-trained weights with frozen base layers and a custom dense classification head.

Results: Validation Accuracy Comparison



Key Findings

Xception achieved the highest accuracy at **97.7%**, demonstrating strong generalization with depthwise convolutions.

MobileNetV2 delivered **96.0%** with only 3.5M parameters — 6.5× lighter than Xception — making it ideal for resource-constrained deployment.

ResNet50 underperformed at **76.0%** due to severe overfitting on the relatively small dataset.

Real-World Testing & Deployment

Real-World Validation

Xception correctly identified Aloe Vera (73.10% confidence) and Tulsi (69.92% confidence) from natural-environment photos with varying backgrounds and lighting.

Streamlit Web Application

The top-performing Xception model was deployed via a lightweight Streamlit interface. Users upload leaf images through a browser for instant species identification.

Google Gemini Integration

A conversational AI assistant provides medicinal properties, active compounds, usage tips, and precautions — transforming the tool into a comprehensive learning platform.

Conclusion & Future Directions

Best Model: Xception

97.7% validation accuracy via depthwise separable convolutions — ideal for fine-grained leaf classification.

Mobile Deployment Ready

MobileNetV2's 96% accuracy at 3.5M params makes it optimal for field and edge-device use.

Future Roadmap

Expand dataset diversity, add multi-language support, enable offline field mode, integrate geolocation mapping, and explore ensemble methods for improved accuracy.

This work demonstrates how AI bridges traditional botanical knowledge and modern technology — enabling intelligent, affordable, and accessible medicinal plant identification for health, research, and sustainability.